

PLC: CPU 1214C DC/DC/DC 6ES7 214-1AG40-0XB0

HMI: KTP700 Basic PN

Sequence of Operations

1. Operator presses the Start Process button on the HMI.
2. PLC checks interlocks: E-Stop is NOT pressed, No Pump Overload faults, and Tank is NOT full (MainTank_Level_2 is FALSE).
3. PLC energizes MainTank_LiqA_Valve and Pump1_Run. Liquid A begins to fill the tank.
4. When MainTank_Level_1 is reached, PLC closes Valve A and stops Pump 1.
5. PLC opens MainTank_LiqB_Valve. Liquid B fills the tank (via gravity or a separate supply).
6. When MainTank_Level_2 is reached, PLC closes Valve B.
7. PLC turns ON the MainTank_Heater.
8. When the MainTank_Temp_AI reaches the target temperature set on the HMI, the Heater turns OFF.
9. PLC turns ON the MainTank_Mixer and starts a timer based on the HMI_Mix_Time_Setpoint.
10. When the mixing time is complete, the Mixer turns OFF, and the MainTank_Drain_Valve opens.
11. When MainTank_Level_Low is detected (tank is empty), the Drain Valve closes. The batch cycle is complete and ready for the next start.